

must change as soon as the fireball hits him, to a mixture of green and yellow. This technique is known as *blending* in Python. You can find the mixture by simply subtracting the two colours. In this case, just subtract (0,255,0) and (255,255,0) and the result is (255,0,0)—red.

Next: the fireball will spread on the character's body, over time—so the intensity of the resultant colour (mixture) should also become brighter over time. Here's the role of the blending factor, the value of which lies between 0 and 1. You multiply the blended colour with a blending factor, the value of which increases over time, from 0 to 1. As a result, the intensity of the resultant colour will also increase from 0 to 255.

Keyboard input

Getting user input is a necessity not only for game-play, but also to configure the game preferences. In the previous part of this series, we handled various mouse and keyboard events; we will get user inputs via the same events, so you may want to re-visit them.

Every key on the keyboard has a key constant for identification; in Pygame, it's `K_a` to `K_z` for the alphabets. We don't have anything like `K_A` because upper-case results from combining the Shift key—and the `mod` value (see the previous article) of the key event tells us about the combination state. Let's use these constants to get user inputs—try out the following script.

```
#keyinput.py
import pygame
from pygame.locals import *
from sys import exit

pygame.init()
screen=pygame.display.set_mode((640,480),0,24)
pygame.display.set_caption("Key Press Test")
f1=pygame.font.SysFont("comicansms",24)

while True:
    for i in pygame.event.get():
        if i.type==QUIT:
            exit()
        a=100
        screen.fill((255,255,255))
        if pygame.key.get_focused(): #1
            press=pygame.key.get_pressed() #2
            for i in xrange(0, len(press)):
                if press[i]==1:
                    name=pygame.key.name(i) #3
                    text=f1.render(name, True,(0,0,0))
                    screen.blit(text,(100, a))
```

```
a=a+100
pygame.display.update()
```

After running the script, press some keys, and you will see the names of the pressed keys on a white screen. I pressed <Left-Ctrl> <Left-Shift> <a> and got Figure 2 as my result.

There are three important functions used in the above script:

- `pygame.key.get_focused()`: The Pygame window will handle the key events only if the window is focused. It returns `True` if the window is focused, and `False` if not.
- `pygame.key.get_pressed()`: Returns a list of Boolean values, one for every key. It returns 1 if the key is pressed, and 0 if not.
- `pygame.key.name(argument)`: It returns the key name from the key constant value; for example, it returns 'space' given 'K_space' as the argument. You can get the values easily by looking for `pygame.locals` in `pydoc` (run `pydoc pygame.locals` in a terminal).

There are many more key functions like `get_mods()`, `set_mods()`, etc, which you can research on the Web. I am pretty sure that after understanding the above script, you will have no trouble getting user input.

Go Online to learn Linux Device Drivers



Core skills like kernel programming or Linux device drivers should be learned at a slow and steady pace for fully comprehending it, says Mr. Raghu Bharadwaj, a lead trainer on core Linux programming. Unlike normal application software learning where it is mostly restricted to functionality based learning, system programming involves deep understanding and analysis of issues like the Linux kernel and its subsystems. Your options to learn these skills are classroom based training, learning through experience, self learning through trial and error and through an effective online course.

While all other modes of training has its own merits and demerits, online learning is definitely a step ahead, especially for courses like this. Online courses give you the comfort of accessing it 24x7 and lets you learn things during hours of focus and interest. Another important aspect is that it permits you to go through each session any number of times and fully understand each and every topic effectively.

Mr. Raghu Bharadwaj is a lead trainer at Veda Solutions, you can listen to his **training videos** on Linux kernel and device drivers at www.techveda.org

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